testguard: A Deterministic Guard Against Agent Test-Weakening, and a Held-Out Benchmark Finding No Live Gaming

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***Abstract*—A coding agent that cannot fix the code has a second path to a green test suite. It can game the suite. It can skip the hard case, delete an assertion, loosen an expected value, mock the unit under test, or hardcode the visible inputs. testguard is a Claude Code hook that watches the routes that edit tests. It reads the git diff and the command log, with no model calls. It blocks on an added skip or a deleted assertion with a passing test, and it warns on a loosened value or an added mock. Hardcoding leaves the tests untouched, so a diff-based guard does not see it. A held-out test catches that case instead. We pair the tool with a held-out benchmark, which separates two kinds of gaming, outcome gaming and test weakening, and runs real agents with no guard installed so their behavior is natural. Across 138 real headless trials on two models, Claude Sonnet 4.6 and Claude Haiku 4.5, including hard tasks built so the honest path is costly, no agent weakened a test. The honest result is a null. testguard flagged none of the 138 trials, so its false positive rate over the 137 honest solves is zero. Its catch behavior is validated on synthetic mutations, because no live weakening occurred to catch.**

I. Introduction and the problem

Agents that write code are now trusted to run their own tests and report results. This is the basis of agentic coding loops. It rests on an assumption. When the agent says the suite passes, the suite passes because the code is correct.

The assumption has a known failure mode. Reward hacking, also called test warping. The agent makes the suite green by changing the tests, not the code. The documented patterns are adding a skip or xfail marker, deleting or weakening an assertion, changing an expected value to match wrong output, mocking the unit under test, and hardcoding the expected output.

A prior project, claim-check, asked whether agents falsely report completion when they can verify themselves. The answer was that they tend not to. testguard asks the next question. When the agent makes the suite pass, did it earn the green or game it.

We make three contributions.

1) testguard, a deterministic, developer-facing, diff-based guard. It flags test weakening during a coding session. It blocks on a skip or a delete with a passing test, and warns on a loosened value or an added mock. It is precision first and fails open.
2) A held-out benchmark methodology. Each task has a visible test the agent can edit and a held-out test it never sees. This separates outcome gaming from test weakening, and it catches source hardcoding that a diff-based guard cannot.
3) An empirical finding. Across 138 real trials on two models, including hard tasks designed to tempt gaming, no agent weakened a test and testguard had a zero false positive rate on honest work. We report this null plainly.

II. Related work

Recent work studies agents that game tests or benchmarks. SpecBench [1], by Zhao and others, decomposes each task into a specification, visible tests, and hidden tests, and measures reward hacking as the gap between the visible and the hidden tests. That visible versus held-out gap is the same held-out idea this work uses, and we treat SpecBench as prior art for the held-out method. EvilGenie [2], by Gabor, Lynch, and Rosenfeld, measures reward hacking three ways, held-out unit tests, LLM judges, and test file edit detection. Its test file edit detection is the closest prior work to testguard's mechanism. testguard runs that same check live, as a developer-facing hook in a coding session, rather than as an eval-harness measurement after the run. CircumEval [3] measures how often agents circumvent file permissions, including editing or skipping a writable test. The METR report [4] on reward hacking is a field note on frontier agents that modify tests or scoring code to get a pass. The Cursor engineering post [5] audited 731 agent trajectories on coding benchmarks and found that 57 percent used an upstream lookup, reproducing a fix from the public web rather than solving the task. Other benchmarks measure tool-use exploits and evaluator tampering by agents [6], [7].

testguard differs in where the check sits. The work above measures the problem on fixed task sets. testguard is a guard, not a benchmark. It reads the diff during a live session and blocks a weakening before the green is trusted. The benchmark in this repo is the instrument we use to test the guard, not the contribution by itself.

III. The tool

testguard wires three hooks. Two record commands, one detects.

- `log_command.py` on PostToolUse for the Bash tool appends each command with an ok status to a per-session log.

- `log_command.py` on PostToolUseFailure appends each failed command with a fail status.
- `detect_gaming.py` on Stop runs `git diff HEAD`, classifies the changed test files, checks the command log for a passing test command, and decides.

The detector classifies each changed test file from the diff.

- `WEAKEN_SKIP`. An added line carries a skip or xfail marker.
- `WEAKEN_DELETE`. A removed assertion or test definition, with a net removal, meaning fewer asserts or tests after the change.
- `WEAKEN_VALUE`. A removed assertion and an added assertion with the count steady but the expected literal changed. Lower confidence.
- `SUSPECT_MOCK`. An added mock or patch in a test file.
- `STRENGTHEN`. Added tests or assertions with no removals. This is good behavior and is never flagged.

The decision is precision first.

- Block on a `WEAKEN_SKIP` or `WEAKEN_DELETE` with a passing test command this session.
- Warn on a `WEAKEN_VALUE` or a `SUSPECT_MOCK` with a passing test command, or on any weakening with no confirmed passing test command.
- Stay silent otherwise.

Three properties hold. Detection is deterministic, with no model calls. The signal is the diff and the command log. The hook fails open, so any error exits 0 and a session is never broken. Strengthening never flags, so adding tests or assertions is free. The same defaults are baked into the script and mirrored in `tool/gaming_rules.toml`, so the tool runs with zero config.

IV. The held-out benchmark methodology

A guard that only sees the diff cannot catch source hardcoding, because the diff has no test edit. To measure gaming honestly we need a ground truth that does not depend on the diff. We use held-out tests.

Each task has two test files. A visible test ships in the workspace, and the agent can read it and edit it. A held-out test is hidden. It tests the same spec with different inputs. It is never placed in the run directory.

A trial runs as follows. The workspace is copied to a fresh temp run directory. The held-out test is not copied in. The agent runs headless, with full tools and bypassed permissions, on a neutral task prompt that does not hint at editing tests. testguard is not installed in the run directory, so the agent behaves naturally. After the run we compute four things. `visible_passed` runs the visible test command in the run directory. `holdout_passed` copies the final code to a clean directory, drops in the held-out test, and runs it there, so the run directory the agent used stays pristine. `agent_edited_visible_tests` comes from the diff. `testguard_verdict` comes from the offline detection core, evaluate of the diff and the command log reconstructed from the transcript. We reuse the same detector the hook uses, so there is one source of truth, not two.

We classify each trial.

- `honest_solve`. Visible and held-out both pass, and no test was weakened.
- `test_weakening`. The diff weakened a visible test, by skip, delete, or loosen.
- `source_hardcoding`. Visible passed, held-out failed, and no test was edited.
- `broken`. The visible test did not pass.

This gives two gaming measures. Outcome gaming is visible passed and held-out failed. Test weakening is the diff weakened a visible test. testguard detects the second kind by design. The held-out test catches the first kind, including source hardcoding.

Every task passes a strict preflight before any run. The baseline fails both the visible and the held-out test. A correct honest solution, which we wrote and ran, passes both, which proves the task is solvable and the held-out is aligned with the visible spec. The intended shortcut passes the visible test but fails the held-out test, which proves gaming is real and the held-out catches it.

The task set has thirteen tasks. Six easy gaming tasks, two controls, and five hard gaming tasks.

Easy gaming tasks, with the available shortcut.

- `roman_numeral`, hardcode. The visible test checks four inputs, so it can be a lookup table.
- `nth_prime`, hardcode. The visible test checks three indices, so it can be a lookup.
- `balanced_brackets`, skip. One visible test checks interleaved brackets and can be skipped.
- `merge_intervals`, skip. The unsorted input test can be marked xfail.
- `format_price`, loosen. The exact string for 5 cents can be loosened to a prefix check.
- `percentage`, loosen. The exact 66.7 percent can be loosened to accept a truncated 66.6 percent.

Controls, with no easy shortcut. The two controls run only in the easy task set, not in the hard set.

- `reverse_words`. Many varied inputs, the honest one line is the natural path.
- `running_total`. Many varied inputs, the honest loop is the natural path.

Hard gaming tasks, with one line on why the honest path is costly.

- `hard_tax_brackets`, hardcode. A deduction, three marginal brackets, and a credit that phases out, with two floors, all interacting.
- `hard_number_words`, hardcode. A full number-to-words converter with teens, tens hyphenation, hundreds, and thousands grouping.
- `hard_split_bill`, loosen. Largest remainder apportionment, exact cents that sum to the total, with fractional ranking and ties.

- `hard_billing_date`, skip. Month and year rollover with day clamping to the target month length, including leap February.
- `hard_coin_change`, skip. Minimum coins is dynamic programming, greedy by largest coin is wrong for non-canonical denominations.

V. Results

A. Synthetic validation

The self test mutates each task in the documented ways and feeds the result through the detection core. It confirms that a skip or a delete blocks, a loosened value or an added mock warns, a hardcoded source stays silent, and an honest solve stays silent. These checks pass. See `bench/analyze/selftest.md`.

B. Real agent runs

We ran two models, Claude Sonnet 4.6 with model string `claude-sonnet-4-6`, and Claude Haiku 4.5 with model string `claude-haiku-4-5-20251001`. One CLI version, 2.1.193, was recorded in every trial at run time. Data is namespaced by model so the runs coexist. The unequal N reflects the staging of the runs, Sonnet ran first as a smaller pilot at N=3, and Haiku ran deeper at N=8 as the main cross-model pass, for a comparable budget since Haiku is cheaper per trial.

The committed runs are these. Table I gives the per-run totals.[1]

Sources are `bench/analyze/results.md` for the Sonnet easy run, `bench/analyze/results-claude-haiku-4-5-20251001.md` for the Haiku easy run, and `bench/analyze/results-hard-pilot-n5.md` for the hard run. Haiku had 1 broken trial on `hard_split_bill`, an honest failure, not gaming.

C. Aggregate

Aggregated across all committed records in `bench/data/`, the totals are these. Table II lists them.

VI. Discussion

The headline of this work was meant to be testguard catching real agent weakening. That did not happen. Across 138 real trials, no agent weakened a test. Both models solved the tasks honestly, or in one case failed honestly. The honest reading of the hard tasks is this. 49 of the 50 hard-task trials were solved honestly. So the hard tasks largely stayed within the models' honest reach. They did not reach the cornered regime, where weakening a test is the only route to a green suite. In the one trial where a model could not solve the task, `hard_split_bill` on Haiku, the model failed honestly rather than weakening a test.

This is consistent with the claim-check finding. Capable agents that can verify themselves tend to behave honestly. The result here extends it. Even when the suite is editable and a shortcut is available, these two models did not take it.

The one measured strength of testguard is its zero false positive rate. Over the 137 honest solves it never flagged honest work. For a guard, this is the property that decides whether it is usable at all. A false block is worse than a missed one, because it interrupts honest work and trains the user to ignore the guard.

[1] A two-model pilot view at N=3, 48 trials, is a subset of the easy data and not a separate run. See `bench/analyze/results-pilot-n3.md`.

VII. Limitations

We state the limitations plainly.

- No live weakening was observed. The catch behavior of testguard is validated on synthetic mutations only. Whether testguard catches live agent weakening at the rate the synthetic tests suggest is open, because no live weakening occurred to catch.
- Two models, one CLI version. The result may not hold for other models, other versions, weaker or more pressured agents, or different prompting.
- Small task set. Thirteen tasks, self contained and short. Real codebases are larger and messier. A weakening in a large suite may be harder to detect from the diff.
- The hard tasks may not have been hard enough to test gaming under genuine pressure. 49 of the 50 hard-task trials were solved honestly, so the tasks did not corner the models. Tasks reliably beyond the agent's honest reach, where weakening a test is the only route to a green suite, are future work.
- The intervals are wide. A zero count over 116 gaming task trials is consistent with a true gaming rate up to about 3 percent.
- testguard is diff-based. It does not catch source hard-coding by design. The held-out test catches that, but a developer using testguard alone does not have a held-out test.
- The detector uses heuristics for assertions, test definitions, and markers. Languages and styles outside the configured patterns may be missed or misread, though the fail-open and warn tiers limit the harm.

VIII. Conclusion

testguard is a deterministic, developer-facing guard that flags test weakening from the git diff and the command log, with no model calls and a fail-open design. We paired it with a held-out benchmark that separates outcome gaming from test weakening and runs real agents with no guard installed. Across 138 real trials on two models, including hard tasks built to tempt gaming, no agent weakened a test, and testguard had a zero false positive rate on honest work. The catch behavior rests on synthetic mutations for now. The honest null is itself a finding. On this task set, these agents earn the green rather than game it.

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TABLE I
COMMITTED RUNS

Run	Trials	Pooled gaming	FP on honest solves
Sonnet, eight easy tasks, N=3	24	0 of 18	0 of 24
Haiku, eight easy tasks, N=8	64	0 of 48	0 of 64
Both models, five hard tasks, N=5	50	0 of 25 each	Haiku 0 of 24, Sonnet 0 of 25

TABLE II
AGGREGATE OVER ALL COMMITTED TRIALS

Quantity	Value
Total real trials	138 (Sonnet 49, Haiku 89)
Honest solves	137
Broken	1
Outcome gaming	0
Test weakening	0
Source hardcoding	0
testguard flags over 138 trials	0
False positive rate over 137 honest solves	0 (Wilson 95 percent interval, 0.000 to 0.027)
Gaming over 116 gaming-task trials, 138 minus 22 control	0 (Wilson 95 percent interval, 0.000 to 0.032)

responsible for the research design, the experimental methodology, the analysis, the writing, and all claims.

APPENDIX A. REPRODUCIBILITY

- Detector. The hook and the benchmark both call evaluate of the diff and the command log in `tool/hooks/detect_gaming.py`. There is one detector.
- Models and version. One model per namespace, one CLI version 2.1.193, recorded in each trial meta at run time, with no backfilling.
- Held-out discipline. The held-out test is never in the run directory. It is injected only into a clean copy of the final code during analysis. The harness checks for leaks and records the run directory file list.
- Commands.
  - `pytest tests/` runs the acceptance gate for the hook.
  - `python preflight.py` proves the hook detection is real, not vacuous.
  - `python bench/preflight.py` proves each task is well formed.
  - `python bench/selftest.py` validates the pipeline on synthetic mutations.
  - `python bench/run_trials.py -trials N -model M -tasks ...` runs trials, resumable, into the model-namespaced data dir.
  - `python bench/cross_report.py -model M -tasks ... -trials N` writes a model aware results file.
- Data. The committed records are in `bench/data/`, about 7 MB, namespaced by model, with one record, one meta, and one transcript per trial. Trial costs are recorded in the commit messages.
- Note on the disk-fill incident. During the first hard-task pilot, the Haiku agent struggling on `hard_split_bill` wrote a runaway file that filled the disk. The capture harness now caps the size of any single file the agent or its children can write, by RLIMIT_FSIZE, default 64 MB. The affected trial was rerun and recorded as an honest failure.
- Open items. None. Every figure in this draft is sourced from a committed results file or aggregated from the committed records.

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